

Two observations on scroll segmentation and ink detection in the Herculaneum papyri

An informal technical note

Diego C. V. · May 2026 (revised July 2026)

Abstract

This note presents two independent, low-cost observations that may be useful during the current phase of the Vesuvius Challenge. Following the June 2026 demonstration of the complete virtual unwrapping and reading of PHerc. 1667 [8], the open feasibility questions — can ink be detected? can a whole scroll be unwrapped? — are essentially answered. The frontier has shifted from proof-of-concept to production at scale: reducing cost and data volume, increasing automation, and improving ink quality where the signal is marginal. Both observations below are framed for that phase and are deliberately positioned as improvements and quality-control aids rather than as competitors to the core segmentation and unwrapping pipeline.

The first observation concerns segmentation quality control: the number of papyrus layers in any transverse cross-section is a global count that a valid segmentation must respect, and can therefore serve as a cheap, label-free detector of sheet-switching errors — a post-hoc filter complementary to the winding-angle methods already in use. The second concerns ink detection under a resource constraint: because at least some Herculaneum inks contain lead, spectral information already latent in existing multi-energy scans may raise ink-to-papyrus contrast without acquiring higher-resolution scans (which multiply data volume roughly eightfold). A minimum viable test is proposed using data already public, and — importantly — reformulated as a statistical discrimination test rather than a clean differential subtraction, to respect how the public volumes are actually reconstructed.

1. Background and current state

1.1 The virtual unwrapping pipeline

The pipeline for recovering text from closed, carbonized Herculaneum scrolls has four main stages: (i) volumetric X-ray micro-CT scanning at synchrotron facilities; (ii) segmentation of the papyrus surface within the 3D volume; (iii) virtual flattening (unwrapping) of the segmented surface; and (iv) machine-learning-based ink detection on the flattened surface. Each stage has its own failure modes. This note addresses stages (ii) and (iv).

Scrolls have been scanned at the Diamond Light Source and at the BM18 beamline of the European Synchrotron Radiation Facility (ESRF), which now reaches pixel sizes of roughly 3–9 μm depending on the acquisition mode; helical multi-pass acquisitions allow a large scroll to be scanned at a few micrometres in a couple of hours [1, 3]. In June 2026 the team reported the first complete virtual unwrapping and reading of a sealed scroll, PHerc. 1667 [8], establishing that end-to-end unwrapping of a suitable scroll is achievable. What remains is to make that result routine, cheaper, and applicable to the several hundred scrolls still unread.

1.2 The segmentation problem, after the 2026 milestone

Segmentation — tracing the individual papyrus layers within the CT volume — was for years the limiting step. It is no longer an open feasibility question, but it remains the hardest part to automate at scale, especially the highly compressed inner windings. Three approaches now anchor the community's practice:

- Spiral / surface fitting — a global parametric model of the deformed scroll is fitted to neural-network predictions of the papyrus location, guaranteeing a single continuous sheet as output [2].

- Fiber tracking — following the papyrus fibres to locate the layer. The productive refinement is that one need not track every fibre: tracking the coarse (“fat”) fibres, which survive distorted regions and often run through a whole sheet to the next kollēsis, is sufficient to fix the layer position.
- Projection to a canonical form — first mapping the deformed layers onto a regular reference geometry common to all scrolls, so that segmentation and unwrapping operate on a normalised object.

The most common failure mode across fitting methods is sheet-switching: the fitted surface jumps from one layer to an adjacent one, producing a geometrically inconsistent mesh. Observation 1 targets exactly this failure, as a cheap check rather than a new tracer.

1.3 The ink detection problem

Carbon-based ink and pyrolysed papyrus cellulose have nearly identical X-ray absorption profiles, so they are hard to separate by conventional absorption CT. The advance that made the challenge possible was that the two materials differ subtly in phase-shift response under X-ray phase-contrast tomography, first demonstrated on rolled papyri by Mocella et al. (2015) [4]. That signal is weak and has not generalised across the collection: models trained on the readable scrolls rely substantially on morphological cracks and bright spots rather than on an ink signal per se, and transfer poorly to other scrolls [3]. Ink-to-noise remains marginal in many regions. This is the gap Observation 2 addresses.

1.4 Ink composition: the metallic finding

The assumption that Herculaneum inks were purely carbon-based has been revised by synchrotron X-ray fluorescence work. Brun et al. (2016) found lead at concentrations too high to be explained by contamination in two fragments from the Institut de France collection [5], and Tack et al. (2016) quantified and characterised the lead by complementary X-ray techniques and Monte-Carlo simulation [6]. Earlier spot analyses had already reported heavy metals including lead and strontium [7].

The composition is not uniform: individual scribes prepared their own inks, and lead was not found in every analysed fragment, so any metallic-contrast method applies only to a subset of scrolls. The status of lead-drawn ruling lines is unsettled — Brun et al. discuss possible ruled lines but conclude that no additional material was used for that purpose in their sample [5] — so a distinct “ruling” signal should not be assumed. What matters for imaging is that where lead is present, it has characteristic absorption edges and a dispersive phase response at accessible X-ray energies, opening a route to spectral discrimination.

2. Observation 1 — Layer count as a segmentation quality-control check

Positioning. This is not a segmentation method; it is a label-free error detector that can be run on the output of any tracer. Its value in the production phase is that it flags sheet-switching automatically, without ground-truth text, at negligible cost.

2.1 The count

Take any plane perpendicular to the scroll axis. The number of distinct papyrus layers N in that cross-section is set by the winding geometry — to first order, for an Archimedean winding of average layer thickness t between an inner radius r_o and an outer radius r_{out} :

$$N \approx (r_{out} - r_o) / t$$

Away from the two ends, N does not vary along the axis. Local crushing, folding or lateral displacement changes the geometry of individual layers but not how many layers are present. So a necessary condition holds: any valid segmentation of the complete surface must recover the same layer count at every interior cross-section.

2.2 Use as an anchored check

- Anchor: search the volume for a cross-section where layers are best separated (e.g. maximising a local gradient or phase-contrast separation metric) and count N there with high confidence.
- Flag: at every other interior cross-section, a candidate segmentation yielding $N \pm 1$ surfaces contains at least one sheet-switching or merge error by construction. This is a deterministic flag that needs no text labels.
- Report: emit a per-axis profile of the recovered count so that reviewers see exactly where a mesh gains or loses a layer.

2.3 Relation to existing methods

Winding-angle methods (e.g. graph-based stitching, spiral fitting) enforce local consistency and a relative ordering of patches [2]. The count is complementary and global: it supplies the absolute number the ordering must sum to. The two combine naturally — connectivity gives the order, the count gives the total — but the count adds most value simply as an automated QA pass over meshes the pipeline already produces.

2.4 Limitations

- A reliable anchor may not exist in uniformly compressed scrolls.
- N is an integer; a ± 1 error at the anchor propagates globally, so the anchor count must be conservative.
- The innermost windings may genuinely present fewer resolvable layers than exist physically; the check must carry a tolerance there and is best read as “where does the count change, and is that change physically plausible?” rather than as a hard reject.

3. Observation 2 — Spectral ink contrast from existing multi-energy scans

Positioning. The resource argument is the point. Improving ink signal by acquiring higher-resolution scans multiplies data volume by roughly eight per added dimension, which the community has identified as a growing cost problem. Where two energies already exist for the same object, the spectral information is a sunk cost waiting to be used.

3.1 Physical basis

In the X-ray regime the complex refractive index of a material is written

$$n = 1 - \delta + i\beta$$

where δ (the refractive-index decrement) governs the phase shift and β the absorption. Computing both from tabulated scattering data for carbon papyrus and lead-loaded ink (accompanying script, `delta_beta_ink.py`) gives a clear and slightly counter-intuitive result that should shape how the idea is pitched. In the phase channel the ratio $\delta_{\text{ink}}/\delta_{\text{papyrus}}$ is nearly flat and weak — only a few percent even for high lead loading — because δ tracks electron density and both materials are carbon-dominated; this is exactly why single-energy phase discrimination is hard. The strong signal is in the absorption channel: lead's K-edge at ≈ 88 keV makes the ratio $\beta_{\text{ink}}/\beta_{\text{papyrus}}$ jump sharply — of order tens- to hundreds-fold across the edge for realistic loadings — which is the classic setting for K-edge subtraction: imaging just below and just above 88 keV, where lead's absorption steps up and carbon's does not. Strontium offers a second, lower edge (≈ 16 keV). The script emits both ratio curves and a table at the acquired energies (54, 70, 88, 105 keV) so the optimal energy pair can be read off directly. The operative reframing is therefore from “differential phase-contrast” toward “K-edge subtraction”, which places the idea on the footing of an established dual-energy method rather than a weak phase effect.

3.2 Why a clean differential is the wrong model — and what to do instead

A tempting formulation is a calibrated subtraction that cancels the papyrus and leaves the ink:

$$D(x, y) = I(x, y, E_1) - k \cdot I(x, y, E_2)$$

with k fixed from the papyrus background. This is the right intuition but the wrong operational model, because the public volumes are not raw phase maps $\theta(x, y, E)$: they are tomographic reconstructions produced after a single-material phase-retrieval step, so a pixel value is not a clean, linearly separable δ . Subtracting two such volumes mixes reconstruction assumptions and tends to amplify registration and noise artefacts rather than isolate lead.

There is also a channel question that decides what the public data can and cannot show. The strong K-edge signal lives in attenuation; the public phase-contrast volumes are produced by single-material phase retrieval, a step that encodes δ and can suppress exactly the β information carrying the lead signal. So a test on existing phase-retrieved volumes probes mainly the weak phase feature, whereas the strong K-edge-subtraction signal generally needs acquisitions that preserve absorption. The pilot below should therefore be read as a lower bound: a null on phase-retrieved volumes does not rule out a strong K-edge result on appropriate data, and that distinction should be stated explicitly in any submission.

The robust alternative is to treat the two energies as two features and ask a statistical question. For a fragment with exposed ink (infrared photography giving ground-truth ink masks), register the two energy volumes, form the per-voxel pair (I_{E_1}, I_{E_2}) , and test whether that joint distribution separates ink from papyrus better than either energy alone — e.g. compare the AUC of a two-feature logistic classifier against each single-energy AUC. No deep learning, no reconstruction of δ . Materials of known composition present in the field of view can serve as a cross-energy calibration baseline. A positive result means an exploitable spectral ink feature; a negative result is still publishable, because it informs the scanning protocol for the remaining scrolls.

3.3 Minimum viable test with public data (resource-bounded)

- Phase 0 — reconnaissance (kilobytes): from the data browser alone, identify the fragment with two energies and the best available registration, confirm whether a surface volume exists at both energies, and locate any known-composition materials in the field of view. Decide everything else from this before downloading imagery.
- Phase 1 — pilot ($\approx 1\text{--}2$ GB, streamed, nothing stored): one $\sim 1\text{ cm}^2$ ink ROI, a thin band around the surface, coarse registration on decimated volumes then refined within the ROI, then the AUC comparison above. Gate: if the joint AUC does not beat the best single energy in the ROI, record the null and stop — no further data is spent.
- Phase 2 — only on a positive gate: extend across the fragment in streaming mode.

3.4 If positive: a targeted full experiment

- Compute $\delta_{\text{ink}}(E)$ and $\delta_{\text{papyrus}}(E)$ from tabulated scattering factors for the measured ink composition and pick the energy or energy pair that maximises contrast (script provided).
- Propose synchrotron time to scan a closed scroll at the two optimal energies in phase-contrast mode.
- Calibrate per scroll, since ink composition varies between scribes, using the outermost partially exposed layers rather than a universal model.

3.5 Limitations and open questions

- Only a subset of scrolls carry metallic ink; the fraction among the unread scrolls is not known.
- A metallic signal need not be text: any lead features (should ruling lines exist in a given scroll) would require morphological filtering to separate from writing.

- High-energy phase-contrast near the Pb K-edge needs longer propagation and adds reconstruction complexity.
- Whether the lead concentration yields a detectable differential above the noise floor is the quantitative question the script and the pilot are meant to settle.
- A parallel multi-spectral and 3D-scanning effort on the open fragments (led from the University of Kentucky) is adjacent to this idea; the present proposal is distinguished by working from existing CT energies rather than new multispectral surface imaging.

4. Proposed next steps, by cost

Step A — immediate, no new data.

- Run the layer-count check (§2) as a post-processing QA pass over existing segmentation outputs and emit per-axis count profiles.
- Validate the author's spatial-coherence metric (separate repository) by controlled-degradation curves on the public Paris-4 surfaces — zero new data — to calibrate how its score falls under rotation, shear and misalignment.
- Carry out Phase 0 reconnaissance for the bi-energy test (§3.3).

Step B — short term, computation only.

- Generate the C and Pb scattering-factor curves and identify the optimal energy pair (script provided).
- Run the bi-energy pilot (§3.3, Phase 1) under its data gate.

Step C — medium term, requires synchrotron access.

- Propose and validate a targeted dual-energy phase-contrast scan on a scroll with partially known content.

5. References

- [1] Vesuvius Challenge team (2023–2025). *CT scans of Herculaneum papyri — open data repository*. scrollprize.org/data
- [2] Henderson, P. (2025). *Virtually Unrolling the Herculaneum Papyri by Diffeomorphic Spiral Fitting*. arXiv:2512.04927; WACV 2026.
- [3] Vesuvius Challenge team (2025). *Master plan: reading the Herculaneum scrolls*. scrollprize.org/master_plan
- [4] Mocella, V., Brun, E., Ferrero, C., Delattre, D. (2015). *Revealing letters in rolled Herculaneum papyri by X-ray phase-contrast imaging*. Nature Communications 6:5895. DOI: 10.1038/ncomms6895
- [5] Brun, E., Cotte, M., Wright, J., Ruat, M., Tack, P., Vincze, L., Ferrero, C., Delattre, D., Mocella, V. (2016). *Revealing metallic ink in Herculaneum papyri*. PNAS 113(14):3751–3754. DOI: 10.1073/pnas.1519958113
- [6] Tack, P., Cotte, M., Bauters, S., Brun, E., et al. (2016). *Tracking the ink composition on Herculaneum papyrus scrolls: identification, localization, quantification and speciation of lead*. Scientific Reports 6:20763. DOI: 10.1038/srep20763
- [7] Seales, W. B. et al. (2009). *Ink identification in Herculaneum papyri: first evidence of heavy metals*. Lire Sans Détruire, Académie des Inscriptions et Belles-Lettres.
- [8] Vesuvius Challenge team (2026). *Complete virtual unwrapping and reading of a rolled Herculaneum papyrus (PHerc. 1667)*. arXiv:2606.29085.

Revision note (July 2026): state of the art updated after the June 2026 complete-unwrapping result; Observation 1 reframed as a quality-control check; Observation 2 reframed on a resource basis and its test restated statistically; citation [5] corrected to Brun et al. (2016) and Tack et al. added as [6]; the ruling-line claim softened. The quantitative scattering-factor prediction is delegated to the accompanying script for reproducibility.